

SK. MD. ASIF NEWAZ

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RESEARCH INTERESTS

Machine Learning · Multimodal Integration · Health-Informatics · Data-Analytics · LLM · RAG · Explainable AI · Medical Image Analysis · Contrastive Learning · Computer Vision · Healthcare AI

EDUCATION

B.Sc. in Computer Science and Engineering September 2021 – April 2026
American International University-Bangladesh (AIUB), Dhaka, Bangladesh
CGPA: 3.89/4.00
Relevant Coursework: Machine Learning (Python), Introduction to Data Science (R), Research Methodology, Artificial Intelligence and Expert System (Python), Computational Statistics And Probability (MATLAB).
Thesis: "Binary Retrieval-Augmented Generation (BinaryRAG): Contrastive Learning and Multimodal Integration for Enhanced Monkeypox Detection"

RESEARCH EXPERIENCE

Research Intern April 2026 – Present
Elite Research Lab PLC. Queens, NY, USA (Remote)
Supervisor: Md Kishor Morol

- Audited multiple datasets and identified criterion feature leakage in a high-visibility medical dataset on kaggle (23.8K downloads)
- Contributed to an RAG pipeline and LLM fine-tuning project, focusing on retrieval optimization and domain adaptation.
- Conducted research in multimodal data integration strategies and data integrity validation protocols to ensure methodologically sound clinical prediction models.

Research Intern March 2025 – June 2025
Applied Intelligence and Informatics Lab (AIIL). Nottingham, United Kingdom (Remote)
Supervisor: Dr. Md. Saef Ullah Miah

- Conducted in-depth literature reviews and developed/fine-tuned large language models, including Mistral-7B-Instruct-v0.3, for culturally aware sales negotiation systems in commercial contexts.
- Compiled datasets, performed model selection and comparison across architectures (Flan-T5, Gemma-2B, LLaMA-3B), and addressed technical implementation challenges in resource-constrained environments.
- Managed editorial works and indexing for ICONIP2024-CCIP conference papers via ProConf platform.

PUBLICATIONS

Journal Submission (Thesis – Proposed for submission)
Binary Retrieval-Augmented Generation (BinaryRAG): Contrastive Learning and Multimodal Integration for Enhanced Monkeypox Detection 2026
- Novel Binary-RAG framework combining CLIP visual embeddings + SentenceTransformer clinical metadata with contrastive multi-head attention
- Achieved AP@5: 0.8525 (+24.9%) precision gain on MSLD v2.0 + 25k synthetic records

International Conference Papers (All Accepted/Published)

Chlorella vulgaris Image Classification Using EfficientNetV2-S, ResNet50 and InceptionResNetV2 with Grad-CAM Analysis (ICIIR 2025)

- 82.65% accuracy (EfficientNetV2-S) and mIoU 0.884 (InceptionResNetV2) on CIDACC dataset

Improving Diagnosis and Treatment Planning for Heart Diseases Using Virtual Reality Integrated with Explainable AI [Best Paper Award] (ICIIR 2025)

- Integration of Virtual Reality (VR) with Explainable AI (XAI) to substantially improve diagnostic accuracy, treatment planning, and medical education outcomes for cardiovascular diseases

Predicting Type 1 Diabetes Using Machine Learning (ICIDA 2024)

- Compared RF, Gradient Boosting, SVM, Logistic Regression for early risk prediction

Breast Cancer Detection Using Machine Learning Algorithms (ICIDA 2024)

- Hybrid ensemble (SVM, KNN, RF, AdaBoost, PCA, GB, DT) on benchmark datasets

Assessing Consumer Behavior Towards Sustainable Apparel: A Statistical and Machine Learning Approach (ICIDA 2024)

- Regression, ANOVA, correlation analysis (SPSS) on survey data (n=384)

Data-Driven Diagnosis of Monkeypox: Insights from Machine Learning Models

[Best Paper Award] (TEHI 2024)

- Evaluated SVM, DT, RF, LR, XGBoost, LightGBM; 4–8% gain via targeted data cleaning

Conference Papers (Under Review)

The Follicle Count Paradox: Measuring Diagnostic Leakage in AI-Driven PCOS Prediction

(ICCA 2026)

- Identified Rotterdam criterion leakage of the most popular PCOS dataset on Kaggle with 23.9k downloads in terms of ML based binary classification.

An AI-IoT Based Framework for Real-Time Pest Detection and Automated Mitigation

(ICCA 2026)

- An n8n agentic AI based IoT framework for real time pest detection and mitigation using remote field vehicle.

TECHNICAL SKILLS

Language & Frameworks: Python, R, PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers, CLIP, Sentence-Transformers, LangChain

Deep Learning: CNNs, EfficientNet, ResNet, Inception, Vision Transformers, Contrastive Learning, Multi-head Attention, Reranker

Tools & Platforms: Jupyter, Git, LaTeX, Grad-CAM, Docker, n8n, PgVector, SPSS, Linux, AWS/GCP (basic), LM Studio, Ollama, UnSloth

ACADEMIC ACHIEVEMENTS

- Dean's List Honors (all semesters)

- Academic Scholarship for Excellence

- Recipient of Magna Cum Laude (Silver Medal) for overall academic performance

- Acquired two best paper awards at separate international conferences [TEHI 2024, ICIIR 2025]

REFERENCES

Dr. Md. Saef Ullah Miah

Research Lead, Applied Intelligence and Informatics Lab
Director, FST, American International University-Bangladesh
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Additional references available upon request.

Md. Kishor Morol

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